

Internship in Physical Organic Chemistry

Week 3, Part 2: Learning the xTB workflow

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Goals for this part:

- Learn how to generate a 3D structure from a SMILES string
- Run a geometry optimisation and understand what xTB produces
- Read the output file and locate the total energy
- Understand what the output files are and which ones matter
- Run a calculation on an ion and compare its energy to the neutral

1 Overview

Before working on your actual compound, you will learn the xTB workflow on a simple molecule: ethanol (CH3CH2OH). This lets you focus on understanding the tools without worrying about the chemistry at the same time. Everything you do here — generating a geometry, optimising it, reading the output, comparing energies — you will repeat on your flavonoid in Parts 3 and 4.

Open your terminal and run the startup commands before doing anything else.

Windows:

```
conda activate physorg
set PATH=C:\xtb\xtb-6.7.1\bin;%PATH%
```

macOS:

```
conda activate physorg
```

Move into your practice folder:

```
cd week3\practice
```

macOS:

```
cd week3/practice
```

2 Generating a 3D structure from a SMILES string

The SMILES string for ethanol is CCO. Run OpenBabel to convert it into a 3D coordinate file:

```
obabel -: "CCO" --gen3d -o ethanol.xyz
```

The `--gen3d` flag tells OpenBabel to generate a three-dimensional geometry, not just a flat 2D structure. The result is saved to `ethanol.xyz`.

Open `ethanol.xyz` in a text editor (Notepad on Windows, TextEdit on macOS). You should see something like this:

```
9
C      1.048   0.055   0.070
C      2.563   0.061   0.073
O      3.044   0.703   1.245
H      0.664  -0.445  -0.823
H      0.656   1.077   0.097
...
```

The first line tells you there are 9 atoms (2 carbons, 1 oxygen, 6 hydrogens). The second line is blank. Each remaining line gives an element symbol and three coordinates.

Now open `ethanol.xyz` in Chemcraft (Windows) or Avogadro (macOS) and verify that the structure displays correctly as an ethanol molecule with all bonds present.

If Chemcraft or Avogadro shows atoms with no bonds between them, use **Tools** → **Re-build Connectivity** (Chemcraft) or check that the bond display is enabled (Avogadro). XYZ files do not store bond information — the visualiser draws bonds based on atom distances.

3 Running a geometry optimisation

The geometry OpenBabel generated is a rough starting point. The atom positions are approximately correct, but not at the true minimum energy arrangement. A **geometry optimisation** moves the atoms step by step until the forces on them are essentially zero — this is the lowest-energy structure the program can find starting from the given geometry.

Run the optimisation:

```
xTB ethanol.xyz --opt --chrg 0 --uhf 0 --gfn 2 --norestart > opt.out
```

This takes a few minutes. While it runs, the terminal shows nothing — the output is being redirected to `opt.out`. When the prompt reappears, the calculation is finished.

What the flags mean:

- `--opt` — perform a geometry optimisation
- `--chrg 0` — the molecule has no charge (neutral)
- `--uhf 0` — all electrons are paired (closed-shell)

- `--gfn 2` — use the GFN2 approximation method
- `--norestart` — start fresh, ignoring any previous calculation files in this folder

You will use these same flags for every calculation this week, changing only `--chrg` when working with ions.

4 Reading the output file

Open `opt.out` in a text editor. It is long — several hundred lines. You do not need to read all of it. The most important pieces are described below.

Did it finish correctly?

Search for the phrase `normal termination`:

Windows:

```
type opt.out | findstr "normal termination"
```

macOS:

```
grep "normal termination" opt.out
```

If you see `normal termination of xtb`, the calculation completed without errors. If this line is absent, something went wrong — contact your supervisor before continuing.

Where is the total energy?

The total energy is printed near the end of the output. Search for it:

Windows:

```
type opt.out | findstr "TOTAL ENERGY"
```

macOS:

```
grep "TOTAL ENERGY" opt.out
```

You will see a line like:

```
| TOTAL ENERGY          -5.768548976418 Eh    |
```

The number $-5.768\dots$ is the total energy of the optimised ethanol molecule, in **Hartree** (symbol Eh). The Hartree is the atomic unit of energy. A single Hartree is approximately 627.5 kcal/mol — a very large amount of energy. The total energies of molecules are always large negative numbers in Hartree; what matters for chemistry is the **difference** in energy between two structures, not the absolute value.

Why is the energy negative and so large?

xTB calculates the total electronic energy of the molecule — the energy of all the electrons interacting with all the nuclei. This is always negative because the electrons are attracted to (stabilised by) the nuclei. The larger the molecule, the more negative the number. A single water molecule has a total energy around -5 Eh ; your flavonoid will be around -62 Eh . These large numbers are not directly meaningful — only differences between energies of similar structures tell you something chemically useful.

Did the geometry converge?

Scroll through `opt.out` and find the section that starts with `GEOMETRY OPTIMIZATION CONVERGED`. You will see a table showing the gradient norm at each optimisation cycle. The gradient norm measures the forces on the atoms — when it drops below a threshold (typically 10^{-4}), the optimisation has converged and the geometry is a true minimum.

5 The output files

After the optimisation, several files appear in your folder. Here is what each one is:

File	Contents
<code>opt.out</code>	Full xTB output (what you redirected with <code>></code>). Contains energies, convergence information, and warnings.
<code>xtbopt.xyz</code>	The optimised geometry. This is what you pass to the next calculation.
<code>xtbopt.log</code>	The optimisation trajectory — all intermediate geometries, one per cycle. Useful for debugging if something goes wrong.
<code>xtbrestart</code>	A file xTB uses to restart a previous calculation. Always use <code>--norestart</code> to ignore it.
<code>charges</code>	Partial atomic charges from the calculation. Not needed this week.
<code>wbo</code>	Wiberg bond orders — a measure of bond strength. Not needed this week.

The most important file is `xtbopt.xyz`. Open it in your visualiser and compare the structure to the original `ethanol.xyz`. The bond lengths and angles should be slightly different — the optimisation has found a more favourable arrangement.

Never run two calculations in the same folder without renaming `xtbopt.xyz` first. The second calculation will overwrite it and you will lose the first result. Use `copy` (Windows) or `cp` (macOS) to save a named copy before running anything new.

6 Comparing neutral and protonated ethanol

You will now run the same calculation on protonated ethanol ($[\text{CH}_3\text{CH}_2\text{OH}_2]^+$, charge +1) and compare its energy to the neutral. This is a small version of what you will do with your compound in Part 3.

Step 1 — Save the neutral result.

Before doing anything else, rename `xtbopt.xyz` so it is not overwritten:

Windows:

```
copy xtbopt.xyz ethanol_neutral_opt.xyz
```

macOS:

```
cp xtbopt.xyz ethanol_neutral_opt.xyz
```

Note the total energy of the neutral from `opt.out` (the number after `TOTAL ENERGY`). Write it down.

Step 2 — Generate a protonated geometry.

Open `ethanol_neutral_opt.xyz` in Chemcraft (Windows) or Avogadro (macOS) and add a proton to the oxygen atom:

Chemcraft: go to **Edit** → **Add Atom**, select hydrogen, and click near the oxygen. Adjust its position so it looks like a normal O–H bond (~ 1 Å from the oxygen). Save as `ethanol_OH2.xyz` via **File** → **Save As**.

Avogadro: select the Draw tool, choose hydrogen as the element, and click on the oxygen atom to add a hydrogen. Use the Manipulation tool to adjust its position if needed. Save as `ethanol_OH2.xyz` via **File** → **Export** → **Molecule**.

Step 3 — Optimise the cation.

Run the optimisation with charge +1:

```
xtb ethanol_OH2.xyz --opt --chrg 1 --uhf 0 --gfn 2 --norestart > opt_cation.out
```

Check for normal termination and note the total energy.

Step 4 — Compare the energies.

You now have two energies in Hartree: one for neutral ethanol and one for protonated ethanol. These numbers cannot be directly compared because they correspond to different numbers of atoms (the protonated species has one extra hydrogen). This comparison is only meaningful when both structures have **the same atoms** — which is exactly the case in Part 3, where you will compare different protonation sites of the same ion.

The purpose of this exercise was to practise the workflow: add an atom, save the geometry, change the `--chrg` flag, optimise, find the energy. You will repeat these steps on your flavonoid in Part 3, where the comparison will be chemically meaningful.

Record in section 3W-b of your `scratch.tex`:

- the total energy of neutral ethanol (Eh)
- the total energy of protonated ethanol (Eh)
- one sentence explaining why direct energy comparison between neutral and protonated ethanol is not chemically meaningful

You are ready to proceed to Part 3, where you will apply this workflow to your actual compound.